ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN NETWORK ENGINEERING/ IT

CCS2612 DATA COMMUNICATION AND COMPUTER NETWORKS

FINAL EXAMINATION

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 29th NOVEMBER, 2023

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

Study the description below and answer that follow:

QUESTION 1

A computer network is an autonomous group of connected devices and computers that can exchange data and communicate with one another. This lowers the cost and complexity of integrated services while enabling the common use of hardware, software, and data resources. At its core, a computer network is built around the idea of making resources accessible across multiple devices. This includes sharing of hardware (like printers and servers), software (applications), and data (like files and databases).

Components of a computer network

Understanding the essential elements that make up a network is necessary to delve deeper into the study of computer networks.

Servers: These are high-powered computers that store and supply data to other computers (clients).

Clients: Personal computers, smartphones, or other devices that request and use resources provided by servers.

Network Interface Cards (NICs): These allow computers to connect to a network.

Switches and Routers: These serve as the 'traffic cops' guiding data packets to their intended destinations in the most efficient way.

Network Operating System (NOS): It coordinates network activities and facilitates shared resource use.

Each of these components plays a specific role, contributing to the efficiency and effectiveness of a computer network. Understanding them is, therefore, crucial for anyone looking to delve into the world of computer networks.

Answer the following questions:

a) With the use one (1) of practical example, differentiate between peer to peer and client/server networks. (6 marks)

b) Describe any three (3) associated costs that are incurred with implementing a computer network. (6 marks)

c) Discuss four (4) benefits of using computer networks in a business organization. (8 marks)

d) Discuss four (4) challenges of using computer networks in a business organization. (8 marks)

e) Explain any four (4) security measures that can be implemented on a network. (8 marks)

f) Networks signals are usually affected crosstalk and attenuation. (4 marks)

[Total: 40 marks]

SECTION B: Answer any three (3) questions in this section.

QUESTION 2

a) Transport Control Protocol (TCP) and User Datagram Protocol (UDP) are the two common protocols used for transportation of data on a network. Stating the services, discuss three (3) application of each of these protocols in industry. (6 marks)

b) Describe two (2) why the Open System Interconnection (OSI) model was developed. (4 marks)

c) With the use of two (2) examples in either case, differentiate between Infra-Red (IR) and Radio Frequence (RF). (6 marks)

d) The crossover cables and straight-through cables are a type of unshielded pair cables (UTP) cables used for various connections. Describe the types of connections. (4 marks)

[Total: 20 marks]

QUESTION 3

Zanaco, a government owned commercial bank, has decided to subnet their network to create 55 hosts per subnet, in order to suit business organization needs, the quest for expansion. Given that the IP address 210.168.50.0 is to be subnetted. Showing your work clearly, compute and explain the following:

a) What class is the IP address and why? (2 marks)

b) The number of bits that were borrowed to create the subnets (2 marks)

c) Subnet mask (2 marks)

d) Number of subnets (2 marks)

e) Number of Hosts (2 marks)

f) Subnet Addresses (At least 4) (4 marks)

g) Host addresses (Give a range of host addresses each of the 4 subnets) (4 marks)

h) Broadcast addresses (At least 4) (2 marks)

[Total: 20 marks]

QUESTION 4

a) Automatic Private Internet Protocol Addressing (APIPA) or Link local protocol is a Microsoft protocol used on a local network. Explain the need to have this protocol running and the IP address used. (4 marks)

b) Write down the correct interface and configurations for the following:

i) Configuring a switch name (1 mark)
ii) Configuring line console password (1 mark)
iii) Configuring an enable password. (1 mark)
iv) Configuring a line vty password (1 mark)
v) Encrypting console password (1 mark)
vi) Saving configurations (1 mark)
vii) Restart a switch (1 mark)
viii) Configuring a banner. (1 mark)

c) The Domain Name System (DNS) handles the growing number of Internet users. DNS performs an important function on TCP/IP-based networks of converting hostnames to IP addresses. Discuss the three (3) entries used by DNS. (6 marks)

d) Describe the use of Address Resolution Protocol (ARP) on a network. (2 marks)

[Total: 20 marks]

QUESTION 5

a) Working as a network technician, a number of users approached you complaining that they are unable to connect to the Internet. Discuss the troubleshooting process you would undertake to try and restore connectivity. (4 marks)

b) With the use of one (1) example in each case, describe any three (3) the software utilities used in network troubleshooting. (6 marks)

c) IPv6, a successor to IPv4, was published by the IETF (Internet Engineering Task Force) as a draft standard in December 1998, then it was implemented fully in June 2012. Describe three (3) reasons that necessitated the change from IPv4 to IPV6. (6 marks)

d) Explain the importance of Network Address Translation (NAT) on a network. (4 marks)

[Total: 20 marks]